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Education

- Aug. 2021 – **Ph.D. in Artificial Intelligence**, KAIST, Daejeon
Aug. 2025 Advisor: Prof. Jong Chul Ye
- Feb. 2012 – **M.S. in Electrical Engineering**, KAIST, Daejeon
Feb. 2014 Advisor: Prof. Wan Choi
- Mar. 2008 – **B.S. in Media Communication Engineering**, Hanyang University, Seoul
Feb. 2012 *Summa Cum Laude* (GPA: 4.13/4.5)

Work Experience

- Aug. 2026 – **Assistant Professor**, DGIST, Daegu
Present Department of Electrical Engineering and Computer Science
- Sep. 2025 – **Assistant Professor**, Kookmin University (KMU), Seoul
Aug. 2026 Department of Artificial Intelligence
- Feb. 2014 – **Senior Researcher**, Agency for Defense Development (ADD)
Aug. 2021 Wireless communication and network systems for military applications

Research Interests

Generative AI, Trustworthy/Responsible AI, AI for Science, Physical AI

Publications

* Co-first, † Co-corresponding

- Journal **[J1]** Physics-Guided and Fabrication-Aware Inverse Design of Photonic Devices Using Diffusion Models
Dongjin Seo*, **Soobin Um***, Sangbin Lee, Jong Chul Ye, Haejun Chung
ACS Photonics (2026)
- Conference **[C9]** Self-Improving Diffusion Classifiers with Minority Preference Optimization
Hyunsoo Kim, Jungmyung Wi, **Soobin Um**, Donghyun Kim, Suhyun Kim
ECCV 2026
- [C8]** Beyond Generative Priors: Minority Sampling with JEPA-Guided Diffusion
Sol Park, **Soobin Um**
ICML 2026

[C7] Diverse Text-to-Image Generation via Contrastive Noise Optimization
Byungjun Kim*, **Soobin Um***, Jong Chul Ye
ICLR 2026

[C6] DPAIL: Training Diffusion Policy for Adversarial Imitation Learning without Policy Optimization
Yunseon Choi, Minchan Jeong, **Soobin Um**, Kee-Eung Kim
NeurIPS 2025

[C5] Boost-and-Skip: A Simple Guidance-Free Diffusion for Minority Generation
Soobin Um*, Beomsu Kim*, Jong Chul Ye
ICML 2025

[C4] Minority-Focused Text-to-Image Generation via Prompt Optimization
Soobin Um, Jong Chul Ye
CVPR 2025 (Oral presentation, top 0.74%)

[C3] Self-Guided Generation of Minority Samples Using Diffusion Models
Soobin Um, Jong Chul Ye
ECCV 2024

[C2] Don't Play Favorites: Minority Guidance for Diffusion Models
Soobin Um, Suhyeon Lee, Jong Chul Ye
ICLR 2024

[C1] A Fair Generative Model Using LeCam Divergence
Soobin Um, Changho Suh
AAAI 2023 (Oral presentation)

Workshop **[W1]** Physics-guided Optimization of Photonic Structures using Denoising Diffusion Probabilistic Models
Dongjin Seo*, **Soobin Um***, Sangbin Lee, Jong Chul Ye, Haejun Chung
NeurIPS 2024 Workshop (ML4PS)

Preprint **[P1]** MotionCFG: Boosting Motion Dynamics via Semantic Motion Sharpening
Byungjun Kim, **Soobin Um**[†], Jong Chul Ye[†]
NeurIPS 2026 (under review)

Research Grants & Projects

Apr. 2026 – **Regional Innovation Leading Company Development R&D Program:** AI-
Mar. 2028 Based Battery State Diagnosis for Electric Two-Wheelers
(ongoing) Funded by the MSS (TIPA), Republic of Korea (~KRW 534M / ~USD 400K, Sub-
contract PI share: ~KRW 100M)
Role: **Subcontract PI**

Apr. 2026 – **AI S&T Innovation Technology Development Program:** Development of AI for
Dec. 2029 Multi-Scale Drug Response Transfer Prediction Based on Biological World Models
(ongoing) Funded by the MSIT (NRF), Republic of Korea (~KRW 3B / ~USD 2.2M, Co-PI
share: ~KRW 800M)
Joint research with National Cancer Center
Role: **Co-Principal Investigator**

- Oct. 2025 – **Future Defense Innovative Technology Program:** AI-Driven Long-Range
 Mar. 2028 EO/IR/SWIR Sensor Fusion System
 (ongoing) Funded by the MSIT, Republic of Korea (~KRW 400M / ~USD 300K)
 Role: **Principal Investigator**
- Apr. 2025 – **AI Star Fellowship Program:** Embodied AI with Multimodal Foundation World
 Dec. 2030 Model for Robotics
 (ongoing) Funded by the MSIT and IITP, Republic of Korea
 Role: Participating Researcher
- Sep. 2024 – **Basic Science Research Program Grant** on Robust Generative AI
 Aug. 2025 Funded by the NRF, Republic of Korea
 Role: **Principal Investigator**
- Jun. 2023 – **Development of AI-Based X-Ray Computer-Based Training Program**
 Jun. 2024 Field-oriented Technology Development Project for Customs Administration
 Funded by the MSIT and Korea Customs Service
 Role: Student Researcher
- Jan. 2023 – **Development of AI Technology for Personalized Plug-and-Play Explanation**
 Jun. 2023 **and Verification of Explanation**
 Funded by the IITP, Korea government (MSIT)
 Role: Student Researcher
- Sep. 2021 – **Development of a Framework to Analyze, Detect, and Mitigate/Remove**
 Aug. 2022 **Bias in AI Models and Training Data**
 Funded by the IITP, Korea government (MSIT)
 Role: Student Researcher

Patents

- Registered **[PR18]** Apparatus, Method, Computer-Readable Storage Medium and Computer Program for Assigning Dynamic Frequencies in Wireless Network
 Patent No. 10-2212367, Jan. 2021.
- [PR17]** Method and Apparatus for Satisfaction Degree based Weighted Fair Resource Allocation Optimization in Cognitive Radio Wireless Network
 Patent No. 10-2204935, Jan. 2021.
- [PR16]** Apparatus and Method for Controlling Performance of Receiver for Sub-Device in MIMO Cognitive Radio Systems
 Patent No. 10-2192564, Dec. 2020.
- [PR15]** Full Duplex Pair Matching Method for Improving Network Performance in Full Duplex Network Environment
 Patent No. 10-2178266, Nov. 2020.
- [PR14]** Successive-Cancellation Fano Decoding Apparatus and Method for Decoding Using the Same
 Patent No. 10-2158312, Sep. 2020.
- [PR13]** Apparatus and Method for Controlling Channel of Cognitive Radio
 Patent No. 10-2107015, Apr. 2020.

[PR12] Data Convergence Method for Reducing Overhead of Cognitive Radio Networks

Patent No. 10-2085205, Feb. 2020.

[PR11] Method and Apparatus for Selecting Frequency Band in Cognitive Radio Network

Patent No. 10-2042260, Nov. 2019.

[PR10] Method and Apparatus for Allocating Frequency Resource in Cognitive Radio Ad-Hoc Network

Patent No. 10-2039650, Oct. 2019.

[PR9] Apparatus and Method for Scheduling Slots for Communication of Data Packets

Patent No. 10-2038051, Oct. 2019.

[PR8] Apparatus and Method for Constructing Rate-compatible Polar Code

Patent No. 10-1996026, Jun. 2019.

[PR7] Time Mirroring Method and System for Airborne Relay Communications

Patent No. 10-1901616, Sep. 2018.

[PR6] Space-Time Dynamic Spectrum Access Apparatus Combined by Multi-Beam Array Antenna and Time Division Duplexing and Frequency Division Duplexing

Patent No. 10-1873102, Jun. 2018.

[PR5] Radio Set System and Setting Channel Method for the Radio Set System

Patent No. 10-1832971, Feb. 2018.

[PR4] Phased Array Antenna System

Patent No. 10-1773481, Aug. 2017.

[PR3] Radio Apparatus for Sensing Space Frequency Spectrum

Patent No. 10-1764655, Jul. 2017.

[PR2] Polarization tracking system using dual polarization antenna with variable gain attenuator and the control method of the same

Patent No. 10-1747789, Jun. 2017.

[PR1] A Dynamic Spectrum Access Technique based on OFDM for P2P Communication

Patent No. 10-1632267, Jun. 2016.

Honors & Awards

May 2026 Silver Reviewer Award, ICML 2026

Aug. 2020 Bronze Medal, National Defense Science Award, ADD

Aug. 2020 Achievement Award, ADD

Jun. 2018 Outstanding Paper Award, KICS Winter Conference

Feb. 2012 *Summa Cum Laude*, Excellence Award, Hanyang University

Invited Talks

- Sep. 2026 *Guided Diffusion in AI for Science: from Inverse Design to Rare-Instance Synthesis*, the NRF/DFG Joint Workshop on Machine Learning for Urban Environmental Modeling and Flows
- Jul. 2026 *Inclusive Generative Models for Robust Physical AI*, IEIE Summer School
- Aug. 2025 *MinorityPrompt*, Korean AI Association - KAIA Summer Conference
- Aug. 2025 *MinorityPrompt*, Poster Session at KCCV 2025
- Jul. 2025 *MinorityPrompt and BnS: Recent Advances in Diffusion-Based Minority Sample Generation*, Visual AI Lab, Princeton University
- Jul. 2025 *Physics-Guided and Fabrication-Aware Inverse Design Using Diffusion Models*, The McMahon Lab, Cornell University (Co-presented with Dongjin Seo)
- Aug. 2022 *Diffusion Probabilistic Models: A Gentle Introduction*, SpiderCore

Teaching Experience

- Instructor**
 - AI Ethics (Spring 2026, KMU)
 - Data Structures (Spring 2026, KMU) — *Top 1.97% Teaching Eval.*
 - Generative AI (Fall 2025, KMU) — *Top 20% Teaching Eval.*
 - Data Science (Fall 2025, KMU) — *Top 20% Teaching Eval.*
- TA**
 - AI501: Machine Learning for AI (Spring 2023, KAIST)
 - AI619: AI for Medical Imaging (Fall 2022, KAIST, **Head TA**)
 - EE424: Introduction to Optimization (Fall 2021, KAIST)
 - EE210: Probability and Introductory Random Processes (Fall 2013, KAIST)
 - EE321: Communication Engineering (Spring 2013, KAIST)
 - Data Science: Modeling for Prediction (Feb. 2022, Hyundai Motors)
 - Data Science Master Program (Sep. 2021 – Nov. 2021, Hyundai Motors)

Professional Service

- Area Chair ICLR 2027
- Reviewer CVPR (2024–2026), ICLR (2025–2026), ICML 2026, ECCV 2026, NeurIPS 2026,
(Conferences) WACV 2026
- Reviewer IEEE Transactions on Image Processing (TIP), Transactions on Machine Learning
(Journals) Research (TMLR)